

Attention is All You Need Paper

The paper "Attention Is All You Need," published on June 12, 2017, introduced the revolutionary Transformer model, a novel neural network architecture that has significantly impacted the field of natural language processing. Authored by Ashish Vaswani and seven other researchers, this paper proposed an encoder-decoder transformer model that entirely eschews recurrence and convolutions, relying instead on a self-attention mechanism to draw global dependencies between input and output.

The Transformer model's core innovation lies in its "attention is all you need" approach, where it processes entire sequences in parallel, leading to faster training times and improved performance compared to previous sequence-to-sequence models that relied on recurrent neural networks (RNNs) or convolutional neural networks (CNNs). This architecture has become the foundation for many state-of-the-art models in various NLP tasks, including machine translation, text summarization, and question answering.

References:

https://en.wikipedia.org/wiki/Attention_Is_All_You_Need

<https://arxiv.org/abs/1706.03762>

<https://itobos.eu/index.php/news/817-attention-is-all-you-need>

<https://krypticmouse.hashnode.dev/attention-is-all-you-need>